1.5.2

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Question

Find the ratio in which the Y axis divides the line segment joining the points $(6, -4)$ and $(-2, -7)$. Also find the point of intersection.

Theoretical Solution

Given the points,

$$\mathbf{A} = \begin{pmatrix} 6 \\ -4 \end{pmatrix} \mathbf{B} = \begin{pmatrix} -2 \\ -7 \end{pmatrix} \tag{1}$$

Let the vector $\mathbf{P}$ be

$$\mathbf{P} = \begin{pmatrix} 0 \\ y \end{pmatrix} , \tag{2}$$

The points $\mathbf{A}$,$\mathbf{P}$,$\mathbf{B}$ are collinear.

Formulae

The points to be collinear,

$$\text{rank}\begin{pmatrix}\mathbf{P}-\mathbf{A} & \mathbf{B}-\mathbf{A}\end{pmatrix} = 1 \tag{3}$$

$$\tag{4}$$

Theoretical Solution

$$\mathbf{P} - \mathbf{A} = \begin{pmatrix} -6 \\ y+4 \end{pmatrix} \tag{5}$$

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} -8 \\ -3 \end{pmatrix} \tag{6}$$

$$\begin{pmatrix} \mathbf{P} - \mathbf{A} & \mathbf{B} - \mathbf{A} \end{pmatrix} = \begin{pmatrix} -6 & -8 \\ y+4 & -3 \end{pmatrix} \tag{7}$$

Conversion to Row Echelon form, $R_2 \to R_2 + \frac{y+4}{6} R_1$:

$$\begin{pmatrix} -6 & -8 \\ 0 & -3 + \frac{y+4}{6}(-8) \end{pmatrix} \implies \begin{pmatrix} -6 & -8 \\ 0 & \frac{-4y-25}{3} \end{pmatrix}$$

$$\frac{-4y-25}{3} = 0 \implies y = -\frac{25}{4} \tag{8}$$

$$\therefore \mathbf{P} = \begin{pmatrix} 0 \\ -\frac{25}{4} \end{pmatrix}$$

Formulae

Section formula for a vector P which divides the line formed by vectors A and B in the ratio k:1 is given by

$$\mathbf{P} = \frac{k\mathbf{B} + \mathbf{A}}{k+1} \tag{9}$$

$$k\left(\mathbf{P} - \mathbf{B}\right) = \mathbf{A} - \mathbf{P} \tag{10}$$

$$\implies k = \frac{\left(\mathbf{A} - \mathbf{P}\right)^\top \left(\mathbf{P} - \mathbf{B}\right)}{\left\|\mathbf{P} - \mathbf{B}\right\|^2} \tag{11}$$

$$(\mathbf{A}-\mathbf{P})^{\top}(\mathbf{P}-\mathbf{B}) = \begin{pmatrix} 6 & \frac{9}{4} \end{pmatrix} \begin{pmatrix} 2 \\ \frac{3}{4} \end{pmatrix} = \frac{219}{16} \tag{12}$$

$$\left\|\mathbf{P}-\mathbf{B}\right\|^2 = \left(\sqrt{2^2 + \left(\frac{3}{4}\right)^2}\right)^2 = \frac{73}{16} \tag{13}$$

$$k = \frac{\frac{219}{16}}{\frac{73}{16}} \tag{14}$$

$$\implies k = 3 \tag{15}$$

Therefore the ratio in which point **P** divides the line segment joining **A** and **B** is 3:1

C Code

```c
#include <stdio.h>

// Section formula: returns y-coordinate when x=0
float sec_form(int x1, int y1, int x2, int y2, int *m, int *n,
    float *px, float *py) {
    // Compute ratio
    *m = x1;
    *n = -x2;

    // Point of intersection (x=0)
    *px = 0.0;
    *py = (*m * y2 + *n * y1) / (float)(*m + *n);

    return *py;
}
```

```python
import numpy as np
import matplotlib.pyplot as plt

# --- Define points ---
A = np.array([6.0, -4.0])
B = np.array([-2.0, -7.0])

# --- Function to generate a line segment between two points ---
def generate_line_segment(point1, point2, num_points=50):
    line_segment = np.zeros((2, num_points))
    lambda_vals = np.linspace(0, 1, num_points)
    for i in range(num_points):
        temp = point1 + lambda_vals[i] * (point2 - point1)
        line_segment[:, i] = temp.T
    return line_segment
```

Python Code

```python
# --- Generate line AB ---
x_AB = generate_line_segment(A, B)

# --- Plot line AB ---
plt.plot(x_AB[0, :], x_AB[1, :], label='$AB$')

# --- Plot points A and B ---
all_points = np.vstack((A, B)).T
plt.scatter(all_points[0, :], all_points[1, :], color='red')
```

Python Code

```
# --- Add labels ---
point_labels = [f'A {tuple(A)}', f'B {tuple(B)}']
offsets = [(10, 5), (-30, -10)]
for i, txt in enumerate(point_labels):
    plt.annotate(txt,
                 (all_points[0, i], all_points[1, i]),
                 textcoords=offset points,
                 xytext=offsets[i],
                 ha='center')
```

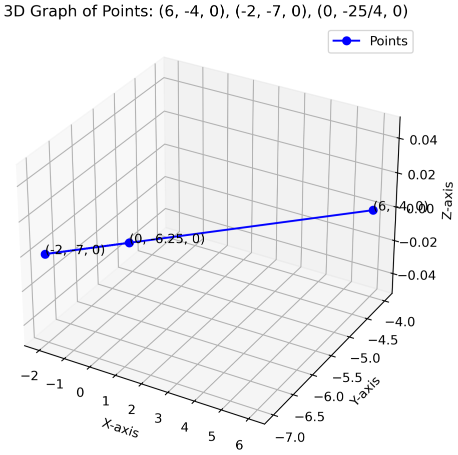
Python Code

```python

# --- Plot settings ---
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.title('Line segment AB')
plt.legend(loc='best')
plt.grid(True)
plt.axis('equal')

# --- Save and show plot ---
plt.savefig('../Figs/graph.png')
plt.show()
```

Plot


3D Graph of Points: (6, -4, 0), (-2, -7, 0), (0, -25/4, 0)
Points
(6, -4, 0)
(0, -6.25, 0)
(-2, -7, 0)
0.04
0.02
0.00
−0.02
−0.04
Z-axis
−4.0
−4.5
−5.0
−5.5
−6.0
−6.5
−7.0
Y-axis
−2
−1
0
1
2
3
4
5
6
X-axis